SLAMpler accurately classifies newly transcribed RNA using a Gibbs Sampler

Austin Szatrowski

aszatrowski@uchicago.edu | GitHub: aszatrowski/SLAMpler

Abstract

Background & the Problem

- SLAM-*seq* measures RNA dynamics by inducing artificial T>C substitutions in newly transcribed RNA
- Low signal: 0.5-2% of Ts substituted + background sequencing error
- Existing method ("GRAND-SLAM") uses two-step estimation: fit global conversion rates with heuristics, then estimate per-gene new RNA fractions (π_g) with EM
- Separating parameter estimation can yield suboptimal inference

Unified Gibbs Sampling Method

- Inspired by STRUCTURE: model sequences as mixtures of population sources with population-specific substitution rates
- Experimentally-derived priors
- Joint estimation via iterative sampling:
  - Assign each read to "new" (high T>C) or "old" (background) RNA
  - Update conversion rates given assignments
  - Update per-gene π_g given rates

Results

- Simulated data (100-2000 reads/gene, π_g = 0.05-1): posterior means recovered true values with proper 95% CI coverage
- Uncertainty scales correctly with read depth
- Substitution rate and individual RNA class estimates noisy but sufficient to determine experimental success

Future directions

- Fully empirical Bayes (derive π_g prior from data) to avoid over-shrinking high-turnover genes
- Automated convergence diagnostics for early stopping

Butterposter Template by Derek Crowe

Figure 1. Posterior tracks true π_g value for all read counts. Lines are posterior means and ribbons are 95% credible intervals. Dashed line is identity (perfect estimation).

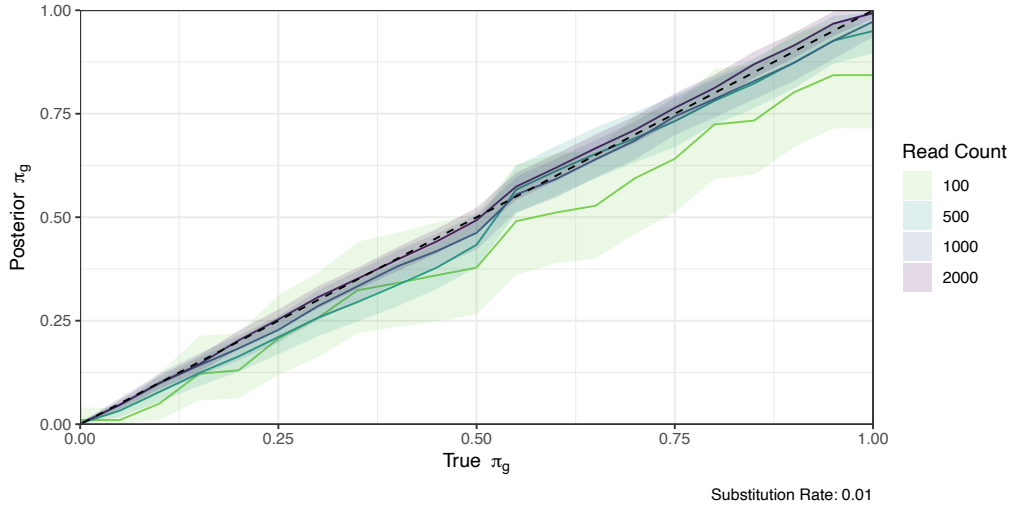

Figure 2. "Worst case": substitution rate of only 0.5%. π_g posterior is noisier.

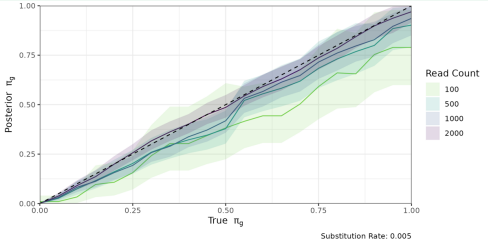

Figure 3. "Best case": substitution rate of 1.5%. π_g posteriors are tighter. Variance shrinks with T>C rate.

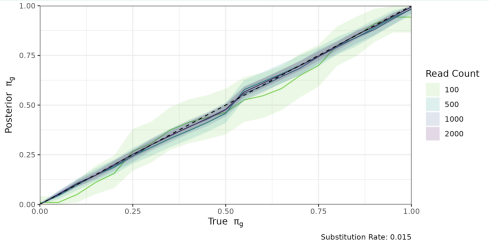

Figure 4. Noisy though fairly accurate estimation of new RNA T>C substitution rate p_n.

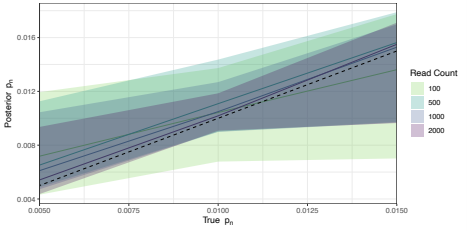
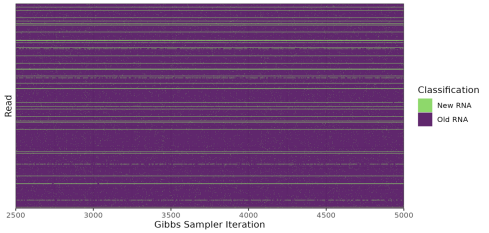
Figure 5. Classifications of individual RNAs over Gibbs iterations post-burn-in. Classes are reasonably stable.